BOOK OF ABSTRACTS

INTERNATIONAL CONFERENCE ON DIGITAL TECHNOLOGIES AND SMART ENGINEERING SYSTEMS (ICDTSES 2026)

FEBRUARY 27th – 28th

ORGANIZED BY
NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY (AN INSTITUTE OF NATIONAL IMPORTANTCE)

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A Sustainable IoT and Machine Learning based Early Warning System for Environmental Safety Monitoring at Waterfall Ecosystems

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Waterfalls function as valuable natural ecosystems yet they create severe environmental and public safety threats because of unexpected water level changes which occur without warning during rainfalls and dam releases and sudden flash floods. The conventional monitoring systems operate through reactive approaches, offer limited support for timely hazard situations and efficient responses. The research demonstrates a Sustainable Intelligent Waterfall Alert System which combines Internet of Things (IoT) sensing infrastructure with hybrid machine learning models to detect hazards before they happen while supporting sustainable environmental monitoring and disaster risk reduction.

The system has the capability to make forecasts on the future hydrological data by leveraging LSTM neural networks and early warning systems that identify patterns regarding the past hydrological data. The system uses the XGBoost classifier, which continuously carries out risk assessments on the present sensor readings and engineered variables, set against the Safe, Warning, and Danger categories. The system avoids false alarms concerning the decision fusion approach taken for the two systems if both decide that the alarm should be raised. The system focuses on three different alarm notification systems that include alerts to people, while the notification systems leverage: RGB LED Signs, Audio Buzzers, and Mobile Dashboard Notification Alerts. The system not only enhances the predictability accuracy and the time taken when making early warnings during pre-testing stages regarding the past and present simulated warnings, but also raises the security level concerning people living around waterfalls since the technique of processing is effective and can thus be easily processed.

Keywords: Internet of Things (IoT), Waterflow Prediction, LSTM, XGBoost, Alert System, Flood Risk, Machine Learning